ABLE ASSESS · CLINICAL BRIEF

Grip strength: the essential biomarker for longevity.

Imagine if a simple handshake could reveal vital insights into someone's healthspan and overall vitality. Grip strength has emerged as a powerful biomarker for biological age — surpassing traditional metrics such as blood pressure in predicting chronic disease risk, health outcomes, and longevity.

AUDIENCE	READING TIME	PREPARED BY
Clinicians & healthcare providers	7 minutes	Able Care · Clinical Team

WHY GRIP STRENGTH MATTERS

A measure of biological resilience, not just physical performance.

Grip strength reflects key health metrics — muscle mass, bone density, and balance — that are critical for maintaining functional independence. As healthcare shifts toward proactive, preventative care, the simplicity and predictive power of grip testing makes a compelling case for its inclusion in routine health assessments.

20 %
HIGHER MORTALITY BY MID-50S
Men with weakest late-teen grip vs.

61 %
HIGHER BREAST CANCER MORTALITY RISK

> BP
STRONGER PREDICTOR THAN BLOOD PRESSURE

1. BIOLOGICAL AGING · 2. MORTALITY

Selective highlights from the evidence base.

01

Marker of biological aging

Lower grip strength correlates with faster DNA methylation age acceleration, a key marker of cellular aging. [Peterson et al., 2022]

02

Indicator of genetic variability

Variations may reflect genomic differences in myotube differentiation and muscle repair, offering insight into personalised aging trajectories. [Matteini et al., 2016]

03

Inflammation & "inflammaging"

Grip strength is a key indicator of inflammation-related risks, linking low strength to higher rates of multimorbidity and age-related decline. [Strasser et al., 2018]

04

Lifespan insights in men

Men with the weakest grip in their late teens are 20% more likely to die by their mid-50s vs. moderate or high grip. [Ortega et al., 2012]

FUNCTIONAL FITNESS & FRAILTY

Preserving independence through early detection.

Grip strength testing identifies functional decline caused by advancing muscle atrophy — a driver of disability and diminished quality of life. Early interventions such as resistance training for 30–60 minutes weekly reduce all-cause mortality, cardiovascular disease and cancer risk by 10–17%. [Momma et al., 2022; Strasser et al., 2018]

KEY INDICATORS ACROSS THE LIFESPAN

01 Early indicator of frailty & sarcopenia

Grip below 44.1 lbs* (men) / 28.8 lbs* (women) flags frailty under the Fried Frailty Phenotype; below 41.0 lbs* / 24.3 lbs* is diagnostic for sarcopenia (EWGSOP). [Fried 2011; Cruz-Jentoft 2019]

02 Cognitive health & dementia risk

Lower grip strength is linked to a nearly two-fold increased risk of cognitive decline and dementia (HR = 1.99). [Cui et al., 2021; Weaver et al., 2002]

03 Cardiovascular & respiratory disease

A grip strength of 62.4 lbs* reduces respiratory disease mortality risk by 51%. Adding grip to risk scores improves prediction of mortality and heart disease. [Cheung 2013; Celis-Morales 2018; Mey 2022]

04 Postmenopausal osteoporosis

Weak grip is associated with reduced bone density and a higher likelihood of falls and fractures — an effective tool for early osteoporosis and fall-risk assessment in postmenopausal women. [Li et al., 2018]

CHRONIC DISEASE

One measurement, many conditions it helps predict.

Grip strength serves as a marker for cardiovascular disease, type 2 diabetes, hypertension, cancer mortality and multimorbidity. Including grip strength in risk scores improves accuracy over traditional metrics alone — making it a high-value addition to routine assessment.

STAT SPOTLIGHTS

"A grip strength of 62.4 lbs* reduces respiratory disease mortality risk by 51%. Adding grip to standard risk scores improves accuracy in predicting mortality and heart disease."

Cardiovascular & respiratory

Cheung 2013 · Celis-Morales 2018 · Mey 2022

Clinical relevance: grip strength is a stronger predictor of cardiovascular events than systolic blood pressure, across all income levels.

"Women aged 40–69 with grip strength below 24.3 lbs* face a 61% higher risk of breast cancer-related mortality — highlighting its value in assessing cancer prognosis."

Cancer mortality in women

Celis-Morales et al., 2018

Clinical relevance: grip strength offers prognostic signal beyond traditional oncology risk assessments, particularly in women 40+.

NOTE ON CUT-OFFS

All cut-off points in this brief have been converted to Able Assess-equivalent measurements for consistency.

GRIP STRENGTH IN PRACTICE

A holistic marker of vitality, built for routine use.

Grip strength enables clinics to create personalised programs that enhance physical and cognitive performance, catch decline early, and reduce downstream healthcare burden.

01

Personalised healthspan

A holistic marker of vitality, enabling clinics to build programs that enhance physical and cognitive performance.

02

Improved QALYs

Early detection of age-related decline preserves functional independence and extends quality-adjusted life years.

03

Reduced healthcare burden

Early identification mitigates the burden of managing chronic conditions, reducing hospitalisations and long-term care needs.

04

Patient empowerment

Shared scores give patients a clear, relatable measure of functional independence they can actively improve.

PRACTICAL CONSIDERATIONS

Use grip strength as a simple screening tool during routine check-ups.

Track trends and assess risk, prioritise early detection of sarcopenia (per EWGSOP), and for high-risk patients provide a hand dynamometer for at-home monitoring. Grip strength is a safer, more practical alternative to gait-speed tests when gait testing may compromise patient safety.

ABLE ASSESS

The only grip strength platform equipped for data-driven healthcare.

Able Assess is a groundbreaking, digitally enabled platform for grip strength measurement. It combines our advanced digital dynamometer with a comprehensive integrative platform and the most up-to-date normative dataset on the market — empowering clinicians with actionable information traditional tools simply can't offer.

- **Understand:** get the essentials on hand dynamometry and how to integrate grip strength into your practice with our comprehensive guide.
- **Explore:** dive into our extensive library of studies on grip strength and longevity.
- **Contribute:** partner with us on research or patient case studies to advance grip strength as a biomarker.

WHAT'S NEXT?

Get in touch to bring Able Assess into your practice.

A member of our clinical team will walk through the platform, the evidence base, and integration options on your schedule.

Get in touch →

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